

consumption patterns, is even more challenging because its successful execution largely depends on local combinations of legislation and infrastructure. Supplying consumers with more detailed information about their energy usage pattern and its effects on energy bills has pushed them to use less energy. A pilot programme conducted recently in Finland where consumers could see the cost of their energy use, resulted in consumers reducing their average energy consumption by 7 per cent.

Solar power, wind turbines and other renewable energy sources are all considered vital contributors to the future electrical energy landscape. Today, however, these are not able to compete commercially with fossil-fuel burning power stations. The challenge is to make renewable-energy sources cost-effective.

This can be achieved by first utilising the accumulated expertise in silicon technology, electronic circuitry and system architectures, to make current renewable sources more cost-effective in producing, storing and distributing energy. Many companies have made good progress in improving the efficiency of solar cells and the electronics

around these, by contributing to the development of better electrical storage technologies that can store energy generated during productive periods when the sun is shining brightly or the wind is blowing hard, and distribute that electricity during less productive times.

Second, we must extend silicon expertise into other semiconductor technologies. New compound semiconductor materials offer better electrical and physical properties than silicon and will increase the efficiency of power transistors. The increase is small (a few percentage points) but it will give the opportunity to develop even more efficient power stages in many applications including hybrid electric vehicles and solar inverters. These materials might also enable new applications driven by a small form-factor power supply. These benefits could be vital in making renewable sources more commercially attractive.

The development and deployment of smart grids also need to be accelerated. Despite amazing technological progress from the late nineteenth century that has dramatically improved most aspects of our lives, there have been few significant changes in the way electricity is gen-

erated and distributed. No doubt upgrading the grid is among the greatest challenges because of the massive but seamless changes required to the electricity infrastructure worldwide. The good news is that there is widespread recognition that smart grids are an essential development. There are many advantages to smart grids, one of which is that these will facilitate the wider deployment of renewable sources.

Ideally, everyone who has invested in renewable resources, for example, by installing solar roof panels or building a small wind turbine, would benefit from a smart-grid upgrade; minimising the power drawn from the grid when these are not self-sufficient in energy and receiving payments when these deliver power back to the grid. Traditional grids provide little incentive for this kind of investment but there is little doubt that these will be an important part of the future energy landscape.

The increasing demand for energy worldwide and the diminishing fossil-fuel reserves are a daunting challenge to the world. Still, companies are working on technologies that give reason to believe that semiconductor technology will contribute greatly to the solutions. ●

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